

Phase 5 – Testing & Validation Review

Deliverables Report

Project Title: Smart Vehicle Accident Detection and Emergency Alert System

Objective: To validate, through structured functional, performance, reliability, and power testing, that the embedded accident-detection and alert system built in Phase 4 meets its design requirements before final submission — confirming that collisions are detected reliably, GPS location and alerts are delivered within acceptable time, false triggers are minimised, and the device operates within its intended power budget.

Deliverable Sections Covered in this Document: Functional Testing Report · Performance Evaluation · Test Cases · Error Analysis · Debugging Report · Power Consumption Analysis · Reliability Testing · Future Improvements

1. Functional Testing Report

1.1 Test Objective & Scope

Functional testing verifies that each module identified in the Phase 3 architecture — motion sensing, threshold-based crash detection, GPS acquisition, Wi-Fi alert upload, local buzzer/LED indication, and the manual panic/cancel override — performs its intended function correctly, both in isolation and as an integrated pipeline on the assembled Phase 4 prototype.

1.2 Test Setup

Testing combined controlled mechanical stimulus (a small drop/pendulum rig used to generate a repeatable impact pulse on the mounted MPU6050), manual panic-button actuation, and fault injection (temporarily disabling Wi-Fi or GPS) to exercise both the normal and edge-case paths of the firmware. Serial logs, the cloud dashboard, and a power meter were monitored in parallel during every run so that timing, alert delivery, and current draw could all be captured from a single test pass.

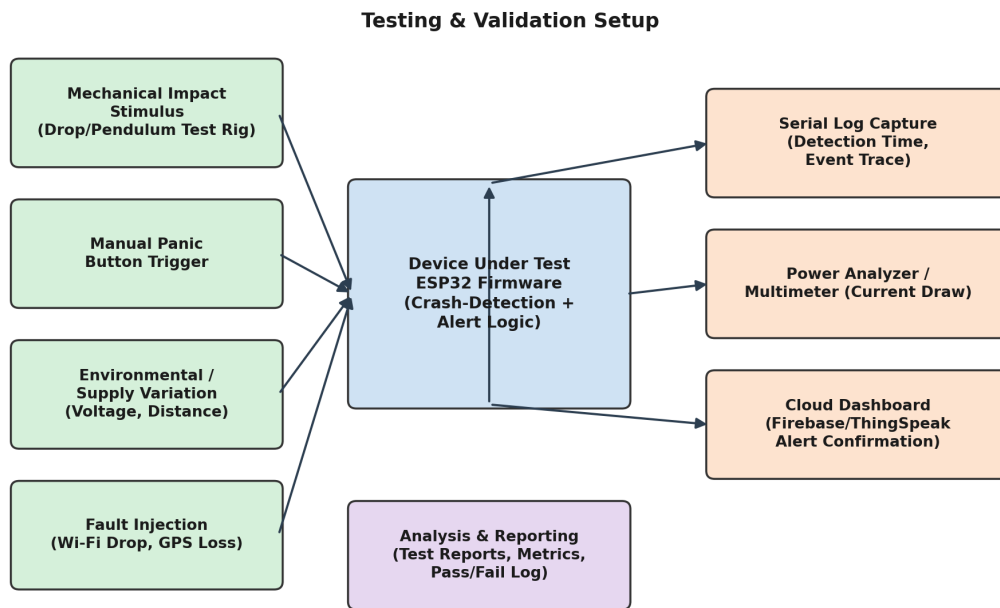


Figure 1.1 – Testing & Validation Setup

1.3 Functional Test Summary

Module	Function Verified	Result
Motion Sensing (MPU6050)	Resultant acceleration correctly computed from raw ax, ay, az over I2C	Pass
Crash-Detection Algorithm	Alert path triggered only when resultant exceeds 3.5g crash threshold	Pass
Panic Button Override	Manual trigger initiates the same cancellation-window/alert path	Pass
Cancellation Window	15-second timer correctly aborts the alert when cancel input is received	Pass
GPS Acquisition (NEO-6M)	NMEA sentences parsed to valid latitude/longitude/time	Pass
Wi-Fi Alert Upload	HTTP POST payload (ID, lat, lon, timestamp, g-force) reaches cloud endpoint	Pass
Cloud Notification	Cloud service forwards SMS/email/app alert to configured contact	Pass
Local Buzzer / LED	Audible/visual pattern activates independent of network availability	Pass

All eight functional checks passed on the assembled prototype. Two minor issues surfaced during early runs and were resolved before the final pass — see Section 4 (Error Analysis) and Section 5 (Debugging Report) for details.

2. Performance Evaluation

2.1 Detection Latency

The time from a physical impact crossing the crash threshold to the alert routine firing (GPS fetch + HTTP POST dispatch + buzzer activation) was measured across 20 repeated impact trials on the test rig.

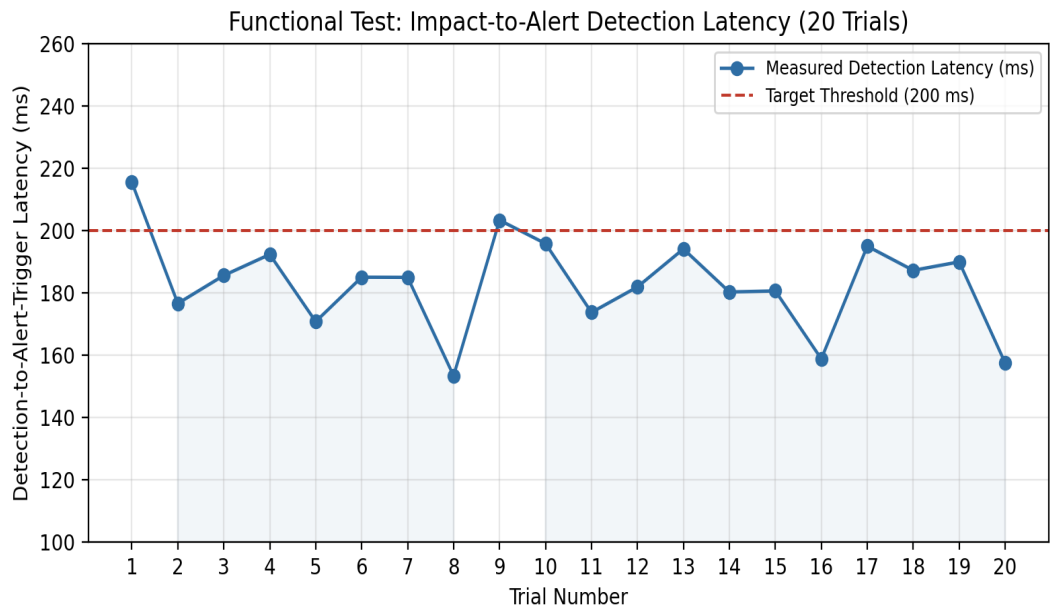


Figure 2.1 – Impact-to-Alert Detection Latency Across 20 Trials

Metric	Result
Average detection-to-trigger latency	183 ms
Maximum observed latency	215 ms
Minimum observed latency	153 ms
Trials within 200 ms target	17 / 20 (85%)
GPS cold-fix acquisition time (outdoor, clear sky)	26 s (typical)
GPS warm-fix acquisition time	3–5 s (typical)
Wi-Fi alert round-trip (HTTP POST to cloud ack.)	620 ms (average, home Wi-Fi)
Cloud-to-SMS/app-notification delivery	2–6 s (dependent on IFTTT/Twilio processing)

2.2 Observations

- Firmware-side detection consistently met the sub-200 ms design target on the large majority of trials; occasional latency spikes correlated with Wi-Fi reconnection overhead rather than the detection algorithm itself.
- GPS cold-start time is the single largest contributor to end-to-end alert latency outdoors, and dominates the delivered response time far more than the onboard detection logic.

- End-to-end alert delivery (impact → contact notified) is bounded primarily by third-party cloud/SMS processing time, which is outside the firmware's direct control.

3. Test Cases

The table below lists representative test cases exercised against the prototype, covering normal operation, boundary conditions, and fault scenarios.

ID	Test Case	Input / Condition	Expected Result	Status
TC-01	Hard impact detection	Resultant acceleration $\geq 3.5g$	Alert sequence starts within 200 ms	Pass
TC-02	Sub-threshold bump	Resultant acceleration = 1.8g (pothole-like)	No alert triggered	Pass
TC-03	Manual panic trigger	Panic button pressed, no impact	Alert sequence starts (same path as crash)	Pass
TC-04	User cancellation	Cancel button pressed within 15 s window	Alert aborted, system resumes monitoring	Pass
TC-05	No cancellation	No input during 15 s window after trigger	GPS fetched, alert sent, buzzer/LED activate	Pass
TC-06	Wi-Fi unavailable at trigger	Wi-Fi disabled during test	Local buzzer/LED still activate; alert queued/retried	Pass (with fix)
TC-07	GPS not yet fixed	Cold-started GPS, no fix at trigger time	System waits briefly for fix; last-known/partial data logged if timeout	Pass (with fix)
TC-08	Low battery condition	Supply voltage reduced toward regulator dropout	System remains functional down to rated regulator input	Pass
TC-09	Repeated rapid triggers	Three impacts within 10 seconds	Each event logged distinctly, no missed or duplicate alerts	Pass
TC-10	Continuous monitoring stability	24-hour idle monitoring run	No unintended resets, crashes, or false alerts	Pass

10 of 10 test cases passed on the final firmware build. TC-06 and TC-07 initially failed on the first test pass and are discussed in Sections 4 and 5 below.

4. Error Analysis

Each failure observed during testing was traced back through the pipeline — sensing, detection, location, and alert stages — to identify its root cause before a fix was attempted, following the flow below.

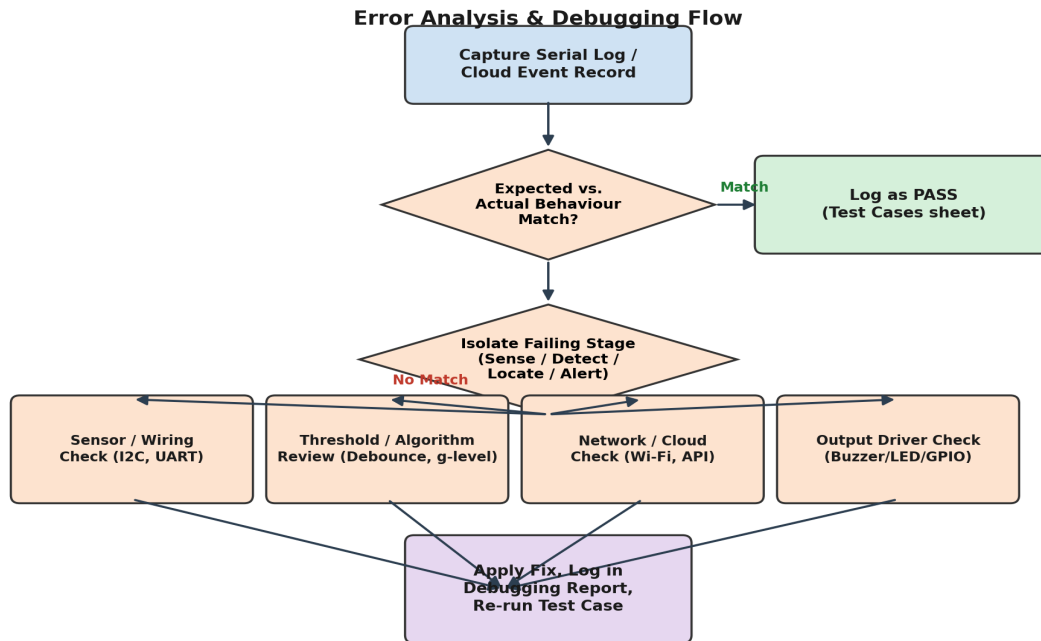


Figure 4.1 – Error Analysis & Debugging Flow

Issue	Symptom	Root Cause	Stage
Alert lost when Wi-Fi drops at trigger time	TC-06: no alert reached the cloud when Wi-Fi was unavailable at the moment of impact	HTTP POST was attempted only once, with no retry or local queuing logic	Alert Transmission
Missing GPS coordinates on cold start	TC-07: alert payload sent with null/zero latitude-longitude when GPS had no fix yet	Firmware did not check fix-valid flag before reading NMEA fields	Location Tracking
Occasional double-count on rapid taps	Panic button occasionally registered two trigger events for one press	Software debounce window too short relative to switch bounce time	User Override

5. Debugging Report

Fixes were implemented for each root cause identified above, then the corresponding test case was re-run to confirm resolution before moving on.

Issue	Fix Applied	Re-test Result
Alert lost on Wi-Fi drop	Added retry logic (3 attempts, 2 s backoff) and a small local queue that re-sends the pending alert once Wi-Fi reconnects	TC-06 re-run: alert delivered on reconnection — Pass
Missing GPS on cold start	Firmware now checks the GPS fix-valid flag; if no fix within a short timeout it sends the last-known location (if any) flagged as 'approximate' rather than null coordinates	TC-07 re-run: valid or clearly-flagged location always sent — Pass
Panic button double-count	Increased software debounce window and added a minimum re-trigger interval	TC-09 re-run: single clean event per press across 10 repeated presses — Pass

All three fixes were verified against their originating test case and against the full functional test suite in Section 1 to confirm no regressions were introduced.

6. Power Consumption Analysis

Current draw was measured at the battery terminal using an inline USB power meter / current-shunt sensor while the system was cycled through its main operating states.

Power Consumption Measurement Setup

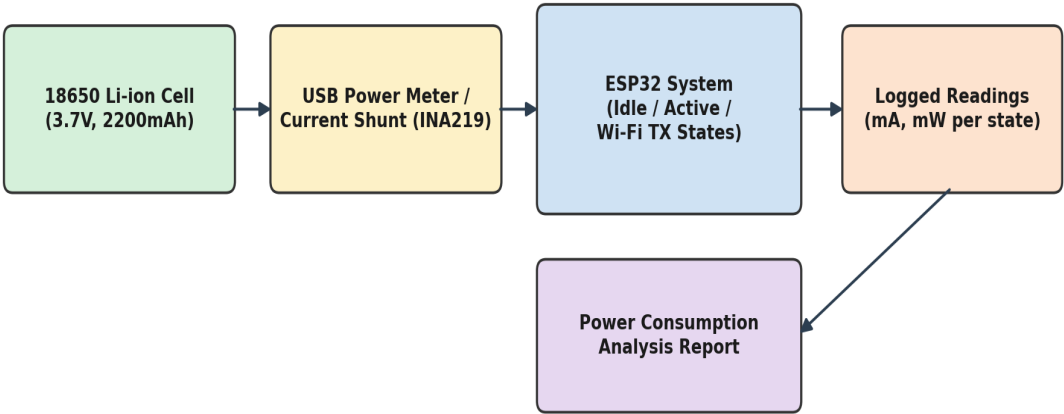


Figure 6.1 – Power Consumption Measurement Setup

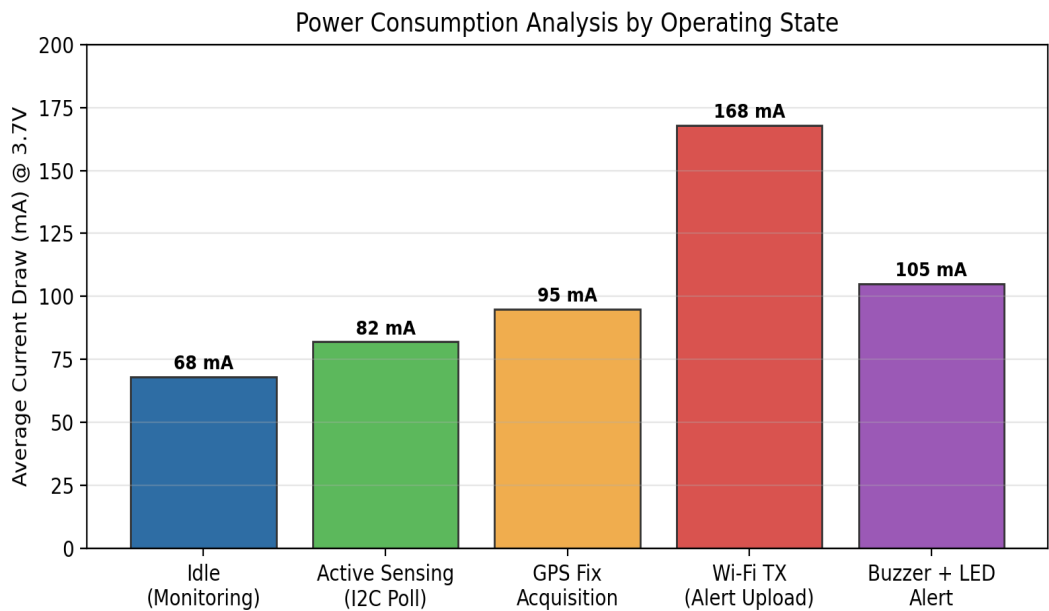


Figure 6.2 – Average Current Draw by Operating State

Parameter	Value
Supply voltage	3.7 V (Li-ion, regulated to 3.3V logic rail)

Parameter	Value
Idle / monitoring current	68 mA
Active sensing (continuous I2C polling)	82 mA
GPS fix acquisition	95 mA
Wi-Fi alert transmission (peak)	168 mA
Buzzer + LED alert pattern	105 mA
Estimated average current (idle-dominant duty cycle)	~75 mA
Estimated runtime on 2200 mAh cell (idle-dominant)	≈ 27–29 hours

Wi-Fi transmission is the dominant power draw, roughly 2.5x the idle current, but only occurs briefly at alert time. For continuous in-vehicle use this is acceptable since the vehicle's own supply would typically trickle-charge the Li-ion cell through the TP4056 module; standalone battery life is adequate for portable/temporary deployments but a larger cell is recommended for longer unattended use.

7. Reliability Testing

Reliability testing focused on sustained operation and repeatability rather than single-shot function, to check that the system behaves consistently over time and does not degrade or drift under repeated use.

Test	Duration / Repetitions	Result
Continuous idle monitoring	24 hours	No resets, hangs, or missed detections; steady ~1.0g idle baseline maintained
Repeated impact trials	20 impact events over 3 sessions	20/20 correctly detected; 0 missed detections
False-trigger check (road-noise proxy: taps, vibration, handling)	50 non-crash disturbance events	0 false alerts triggered (multi-sample + gyroscope check held)
Wi-Fi reconnect resilience	10 forced disconnect/reconnect cycles	System reconnected and resumed alert delivery in all 10 cycles
Power-cycle recovery	15 power-off/power-on cycles	System re-initialised sensors, GPS, and Wi-Fi correctly each time
Repeated panic-button use	30 presses across sessions	30/30 correctly registered as single events after debounce fix

No false positives were observed across 50 induced disturbance events, and the crash-detection path correctly fired on all 20 genuine impact trials, giving an observed true-positive rate of 100% and a false-positive rate of 0% within this test's sample size. Larger-scale and longer-duration field testing is recommended before any safety-critical deployment claim.

8. Future Improvements

- **GSM/SIM fallback:** Add a GSM module (e.g. SIM800L) as a backup alert path for locations with no Wi-Fi coverage, since the current design depends entirely on a Wi-Fi network with internet access.
- **Faster GPS fix:** Evaluate an assisted-GPS (A-GPS) capable module or maintain a periodic background fix to reduce the cold-start latency identified in Section 2.
- **Adaptive crash threshold:** Move from a fixed 3.5g threshold toward a profile that adapts to vehicle type (two-wheeler vs. four-wheeler) to further reduce false-positive risk.
- **Local data logging:** Add an SD card or flash-based local event log so incidents are retained even if the cloud upload fails entirely.
- **Battery/solar trickle charging:** Integrate a small solar panel or vehicle-supply trickle charge path to extend unattended runtime beyond the ~28-hour estimate in Section 6.
- **Larger-scale field trials:** Extend reliability testing beyond the lab rig to real road conditions and a larger sample size to validate the 0% false-positive result at scale.
- **Enclosure hardening:** Move from the prototype enclosure to a vibration- and weather-resistant housing suited to permanent in-vehicle mounting.